src/dynamics.py · coriolis_vector MMA3001 DeepONet project

coriolis_vector

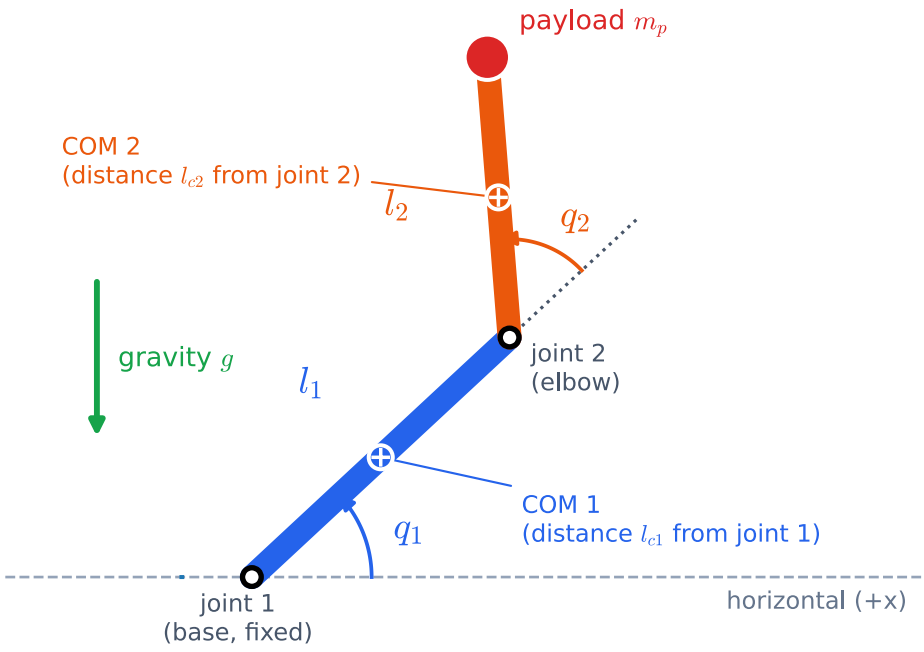
The velocity-dependent torques C(q, q̇)q̇ (Coriolis and centrifugal), derived from the Euler-Lagrange equation

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The set-up: a two-link robot arm in a vertical plane

Everything in this project describes one physical object: a planar robot arm with two rigid links, two motorised joints, bolted to a wall at the origin. It moves in a *vertical* plane, so gravity matters. At the tip of link 2 we may clip on an extra weight (the **payload**). Each joint has a little friction that slows it down in proportion to how fast it spins.



The arm. Blue = link 1, orange = link 2. The angle q_1 is measured from the horizontal; the angle q_2 is measured from the *straight-on extension* of link 1 (dotted), not from the horizontal. The dots with a white cross are the links' centres of mass (COM).

Conventions (fixed everywhere in the project): the state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$; gravity points in the $-y$ direction; potential energy is zero at the height of joint 1; positive angles and torques are anticlockwise.

Symbol	Meaning	Value in the code	Unit
$l_1,\ l_2$	link lengths (joint to joint)	0.5, 0.4	m

$m_1,\ m_2$	link masses (without payload)	2.0, 1.5	kg
$l_{c1},\ l_{c2}$	distance from a link's joint to its own centre of mass (COM)	0.25, 0.2 (middle of each link)	m
$I_1,\ I_2$	moment of inertia of a link about its own COM	$ml^2/12$ (uniform rod)	kg m^2
g	gravitational acceleration	9.81	m/s^2
m_p	point-mass payload at the tip of link 2 (varies per sample)	0 to 1 (dataset), up to 1.5 out-of-range	kg
b	viscous joint friction coefficient (varies per sample)	0.05 to 0.5	N m s/rad
$q_1,\ q_2$	joint angles: q_1 from the horizontal, q_2 relative to link 1	state	rad
$\dot{q}_1,\ \dot{q}_2$	joint angular velocities (a dot means d/dt)	state	rad/s
$\tau_1,\ \tau_2$	motor torques applied at joints 1 and 2 (the inputs)	$A\sin(2\pi f t+\phi)$	N m

Toolbox: Lagrangian mechanics in five minutes

You know Newton's second law for a straight-line motion: $F = ma$. For something that *rotates* about a fixed pin the same law reads

$$\tau = I\,\ddot{\theta},$$

where τ is torque (a twisting push, N m), I the moment of inertia (resistance to being spun, kg m^2) and $\ddot{\theta}$ the angular acceleration. For *one* link that is easy. For *two* links, the second link is moved by the first, which is moved by the base, and the joints push on each other with forces we do not know. Newton's method forces us to compute all those internal forces. **Lagrange's trick** avoids them completely by working with *energies*, which are plain numbers.

The recipe

1. Choose coordinates that describe the pose with no redundancy: here q_1, q_2 (the *generalised coordinates*).
2. Compute the kinetic energy T (as a function of $q, \dot{q}$) and the potential energy V (as a function of q).
3. For every coordinate q_k evaluate the **Euler-Lagrange equation**

$$\underbrace{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right)}_{\text{(a) rate of change of 'momentum'}} - \underbrace{\frac{\partial T}{\partial q_k}}_{\text{(b) pose-dependence of } T} + \underbrace{\frac{\partial V}{\partial q_k}}_{\text{(c) gravity pulls}} = Q_k$$

   where Q_k collects everything we push with from outside (motor torque τ_k, friction).

Reading it: (a) is "how fast is the generalised momentum $\partial T/\partial \dot{q}_k$ changing" (it is the analogue of $d(mv)/dt$); (c) is the force that gravity exerts along coordinate q_k, namely minus the slope of the potential energy (a ball rolls downhill, i.e. the force is $-\partial V/\partial q$, so moving it to the other side of the equation gives $+\partial V/\partial q$); (b) is a correction that appears because the "mass" itself can depend on the pose.

The symbol ∂ means a **partial derivative**: differentiate with respect to one variable while pretending all the others are constants. For instance if $T = \tfrac{1}{2}(3+\cos q_2)\,\dot{q}_1^2$ then $\partial T/\partial \dot{q}_1 = (3+\cos q_2)\dot{q}_1$ and $\partial T/\partial q_2 = -\tfrac{1}{2}\sin q_2\,\dot{q}_1^2$.

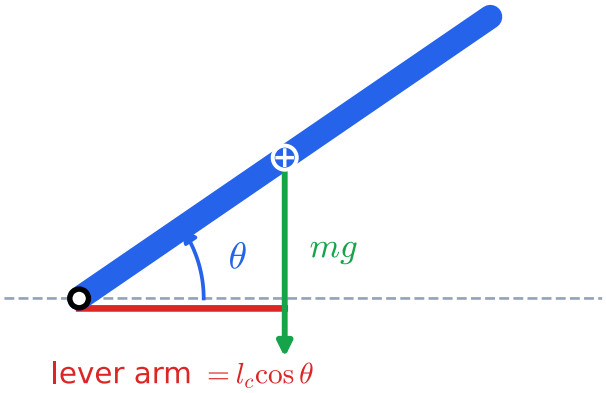
Test run: one swinging rod (you can check it with Newton)

A uniform rod of mass m pivots at the origin, angle θ from the horizontal; its COM is a distance l_c from the pivot. Its inertia about the pivot is $I_p = I_{\text{cm}} + ml_c^2$ (explained in the *combine_payload* document). Then

$$T = \tfrac{1}{2} I_p \dot{\theta}^2, \qquad V = mg\, l_c \sin\theta.$$

Euler-Lagrange: $\frac{d}{dt}(I_p\dot{\theta}) - 0 + mgl_c\cos\theta = \tau$, i.e.

$$I_p\ddot{\theta} + mg\, l_c \cos\theta = \tau.$$



Newton's view of the same rod. Gravity pulls down with force mg at the COM. Only the horizontal distance $l_c\cos\theta$ to the pivot gives a twist (the lever arm), so gravity's torque about the pin is $mg\, l_c \cos\theta$ - exactly the term Lagrange produced.

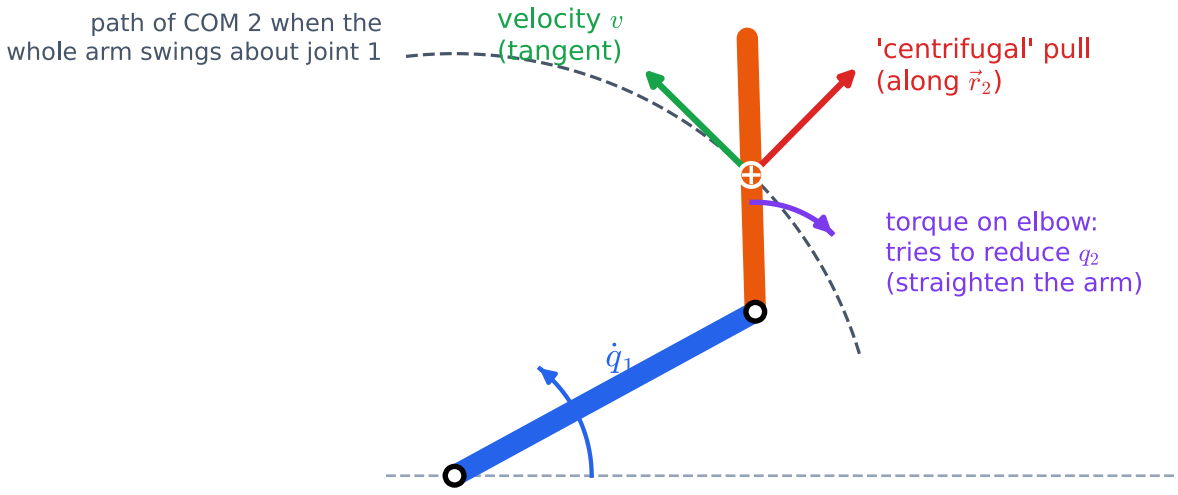
Newton gives $I_p\ddot{\theta} = \tau - (\text{gravity torque})$, the same equation. So the method works, and for two links it needs only energies. That is the whole idea; the rest is careful algebra.

1 The question this function answers

Suppose we apply *zero* motor torque, ignore gravity and friction, and the arm is already moving. A pure "$\tau = M\ddot{q}$" picture says nothing should happen (no torque means no acceleration). But it does: the joint angles keep changing in a curved way, because the arm's inertia changes as it moves. Equivalently, to make the arm follow a prescribed motion, the motors must supply additional torques that depend on the **velocities**. Those are the **Coriolis and centripetal (centrifugal) terms**, written $C(q,\dot{q})\dot{q}$. The function returns this length-2 vector:

```
def coriolis_vector(q, qdot, p):
    q1, q2 = q
    q1d, q2d = qdot
    m2e, lc2e, I2e = combine_payload(p.m2, p.lc2, p.I2, p.l2, p.m_p)
    h = m2e * p.l1 * lc2e * np.sin(q2)
    return np.array([-h*(2*q1d*q2d + q2d**2),
                     h*q1d**2])
```

Everyday analogues: the outward fling you feel in a turning car (centrifugal), and the sideways drift of a ball rolled across a merry-go-round (Coriolis). Both are "inertia wants a straight line, the machine forces a curve" effects.



The physical story behind the term $h\,\dot{q}_1^2$ in joint 2. While the whole arm swings about the base (rate $\dot{q}_1$), the COM of link 2 moves on a circle. Inertia wants it to fly off along the green tangent; the arm holds it on the circle, which is equivalent to an outward pull (red). That pull acts about the elbow and tends to straighten the arm, so the elbow motor must resist it.

2 Where it comes from: the Euler-Lagrange derivative

We start from the kinetic energy found in the *mass_matrix* document,

$$T = \tfrac{1}{2} M_{11}\dot{q}_1^2 + M_{12}\dot{q}_1\dot{q}_2 + \tfrac{1}{2} M_{22}\dot{q}_2^2,$$

where only M_{11} and M_{12} depend on the pose, and only through $\cos q_2$ (with M_{22} constant). Differentiating the cosine gives

$$\frac{\partial M_{11}}{\partial q_2} = -2h, \qquad \frac{\partial M_{12}}{\partial q_2} = -h, \qquad \frac{\partial M_{22}}{\partial q_2} = 0, \qquad \text{where } h := m_{2e}\, l_1\, l_{c2e} \sin q_2.$$

(Check: M_{11} contains $2m_{2e}l_1l_{c2e}\cos q_2$ and $\frac{d}{dq_2}\cos q_2 = -\sin q_2$.) Nothing depends on q_1, so $\partial T/\partial q_1 = 0$. Two more things we need: the chain rule for a time derivative of something that depends on $q_2(t)$, namely $\dot{M}_{11} = \frac{\partial M_{11}}{\partial q_2}\dot{q}_2 = -2h\dot{q}_2$ and $\dot{M}_{12} = -h\dot{q}_2$.

Joint 1

The momentum conjugate to q_1 is $\partial T/\partial \dot{q}_1 = M_{11}\dot{q}_1 + M_{12}\dot{q}_2$. Its time derivative (product rule, since M changes as the arm moves) is

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_1} = M_{11}\ddot{q}_1 + M_{12}\ddot{q}_2 + \underbrace{\dot{M}_{11}\dot{q}_1 + \dot{M}_{12}\dot{q}_2}_{=\,-2h\dot{q}_2\dot{q}_1 - h\dot{q}_2^2}.$$

With $\partial T/\partial q_1 = 0$ the velocity-dependent part of joint 1's equation is therefore

$$\boxed{c_1 = -h\,(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2)}$$

Joint 2

Here $\partial T/\partial \dot{q}_2 = M_{12}\dot{q}_1 + M_{22}\dot{q}_2$, so

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_2} = M_{12}\ddot{q}_1 + M_{22}\ddot{q}_2 + \dot{M}_{12}\dot{q}_1 = M_{12}\ddot{q}_1 + M_{22}\ddot{q}_2 - h\dot{q}_1\dot{q}_2.$$

And this time $\partial T/\partial q_2 \neq 0$: only the pose-dependent parts of T contribute,

$$\frac{\partial T}{\partial q_2} = \tfrac{1}{2}\frac{\partial M_{11}}{\partial q_2}\dot{q}_1^2 + \frac{\partial M_{12}}{\partial q_2}\dot{q}_1\dot{q}_2 = -h\dot{q}_1^2 - h\dot{q}_1\dot{q}_2.$$

Euler-Lagrange takes $\frac{d}{dt}(\dots) - \frac{\partial T}{\partial q_2}$; the velocity terms are

$$c_2 = -h\dot{q}_1\dot{q}_2 - \big(-h\dot{q}_1^2 - h\dot{q}_1\dot{q}_2\big) = \boxed{c_2 = h\,\dot{q}_1^2}$$

(the two $h\dot{q}_1\dot{q}_2$ pieces cancel). These are exactly the two entries returned by the code; the code's minus sign in the first entry and plus sign in the second are the signs we just found.

3 Reading the terms

term	name	story
$-2h\dot{q}_1\dot{q}_2$ in c_1	Coriolis	needs *both* joints moving: swinging the elbow while the whole arm rotates changes how far COM 2 is from the axis, which changes the angular momentum about joint 1.
$-h\dot{q}_2^2$ in c_1	centripetal from joint 2	link 2 spinning about the elbow; its reaction force has a component that twists joint 1.
$+h\dot{q}_1^2$ in c_2	centrifugal from joint 1	the red arrow in the figure: the whole arm swinging flings link 2 outward, torque about the elbow.

> **DOES THE SIGN MAKE SENSE?**
> Sign check with the centrifugal term. The equation of motion reads $M\ddot{q} + c + \cdots = \tau$. With $\tau = 0$ and ignoring the coupling, $\ddot{q}_2 \approx -c_2/M_{22} = -h\dot{q}_1^2/M_{22}$. For $0 < q_2 < \pi$ we have $h > 0$, so $\ddot{q}_2 < 0$: the elbow angle shrinks, the arm **straightens**. That agrees with the picture of the centrifugal pull. ✓

4 The $C(q,\dot{q})\,\dot{q}$ notation

Textbooks write the term as a matrix times the velocity. The matrix that fits ($c = C\dot{q}$) is

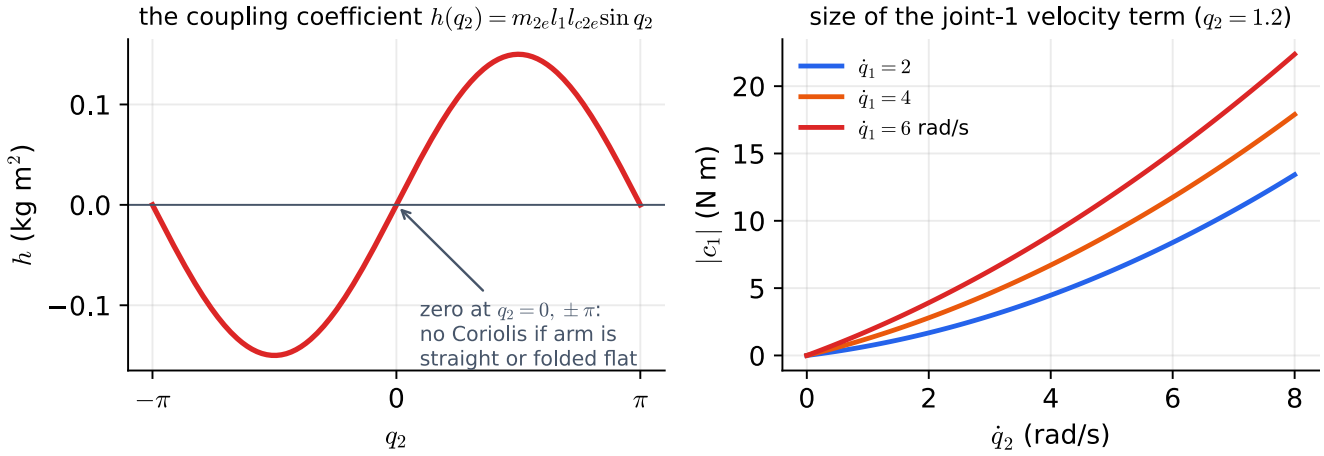
$$C(q,\dot{q}) = h\begin{pmatrix} -\dot{q}_2 & -(\dot{q}_1+\dot{q}_2) \\ \dot{q}_1 & 0 \end{pmatrix} \quad\Longrightarrow\quad C\dot{q} = h\begin{pmatrix} -\dot{q}_2\dot{q}_1 - (\dot{q}_1+\dot{q}_2)\dot{q}_2 \\ \dot{q}_1^2 \end{pmatrix} = \begin{pmatrix} -h(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ h\dot{q}_1^2 \end{pmatrix}.$$

The code skips building C and returns $C\dot{q}$ directly. (A matrix like this has a hidden property, $\dot{M} - 2C$ is skew-symmetric, which is why energy is conserved; see the *energy* document.)

5 Properties

- **Quadratic in velocity.** Double the speeds and every term quadruples (see the plot). At rest ($\dot{q} = 0$) it vanishes.
- **Vanishes for a straight or fully folded arm** ($\sin q_2 = 0$): there the distance of COM 2 from the base is at a maximum or minimum, so a small elbow motion does not change it to first order.

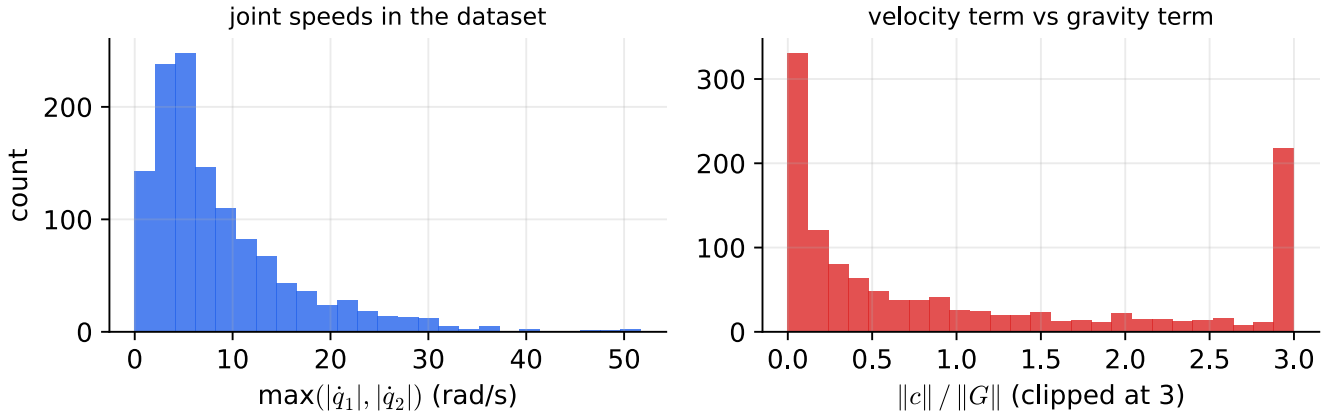
- **Payload enters linearly:** $h = m_{2e}\, l_{c2e}\, l_1 \sin q_2$ and $m_{2e} l_{c2e} = m_2 l_{c2} + m_p l_2$ (see *combine_payload*), so each kilogram of payload adds $m_p l_2 l_1 \sin q_2$ to h.
- **No dependence on q_1.** Like M, only the relative angle matters.
- It does not depend on torque, friction or gravity; it is pure geometry plus inertia.



Left: the coefficient $h(q_2)$, the only place the pose enters. **Right:** magnitude of the joint-1 velocity term $|c_1|$ as the elbow speed $\dot{q}_2$ grows, for three arm speeds $\dot{q}_1$: it grows quadratically.

6 How big is it in our data?

I simulated 40 random dataset trajectories with the real code and measured, at sampled points, the size of c relative to the gravity torque. The joint speeds (95th percentile) are about 24.3 rad/s; the median ratio $\|c\|/\|G\|$ is 0.55 and the 90th percentile 4.79. So the velocity-dependent term is often comparable to, and sometimes larger than, gravity: it is a major reason the dynamics are nonlinear and hard for a neural network to learn.



Left: how fast the joints turn in the project's trajectories. **Right:** the Coriolis/centripetal torque norm divided by the gravity torque norm over those trajectories.

7 Worked example with real numbers

$m_p = 0.5$ kg gives $m_{2e} = 2.000$, $l_{c2e} = 0.250$. Let $q = (0.5,\ 0.8)$ rad and $\dot{q} = (2.0,\ -1.0)$ rad/s.

step	calculation	value
h	$2.000 \cdot 0.5 \cdot 0.250 \cdot \sin 0.8$ ($\sin 0.8 = 0.7174$)	0.17934

c_1	$-h\,(2\cdot 2\cdot(-1)+(-1)^2) = -h\,(-4+1) = 3h$	0.53802
c_2	$h\cdot 2^2 = 4h$	0.71736

These match `coriolis_vector(q, qd, p)`. Individual contributions to c_1: Coriolis $-2h\dot{q}_1\dot{q}_2 = 0.71736$ and centripetal $-h\dot{q}_2^2 = -0.17934$.

8 Code ↔ maths dictionary

code	maths
`h = m2e*p.l1*lc2e*np.sin(q2)`	$h = m_{2e} l_1 l_{c2e} \sin q_2 = -\frac{1}{2}\,\partial M_{11}/\partial q_2$
`-h*(2*q1d*q2d + q2d**2)`	$c_1 = -h(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2)$
`h*q1d**2`	$c_2 = h\dot{q}_1^2$
returned vector	$C(q,\dot{q})\,\dot{q}$

CHECK YOURSELF

1. What is c when the arm is stretched out straight and spinning fast? *Answer:* $q_2 = 0 \Rightarrow h = 0 \Rightarrow c = 0$ (the whole arm acts like one rigid rod).
2. If both joint speeds double, by what factor does c change? *Answer:* 4 (every term is a product of two velocities).
3. $\dot{q}_1 = 3,\ \dot{q}_2 = 0$. Which entry of c is non-zero? *Answer:* only $c_2 = 9h$; c_1 needs $\dot{q}_2 \neq 0$.